The variation of vapour pressure of pure benzene and toluene with its mole fraction is given in the graph.

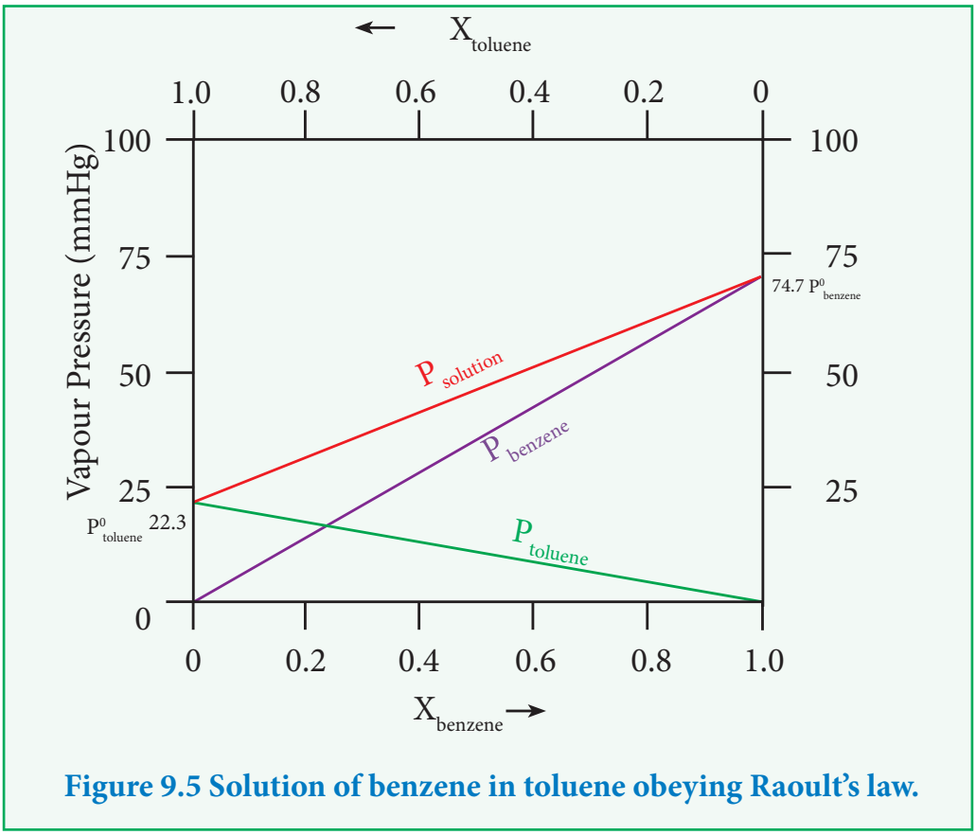

Figure 9.5 Solution of benzene in toluene obeying Raoult's law.

The vapour pressures of pure toluene and pure benzene are 22.3 and 74.7 mmHg, respectively. The above graph shows, the partial vapour pressure of the pure components increases linearly with the increase in the mole fraction of the respective components. The total pressure at any composition of the solute and solvent is given by the following straight line (represented as red line) equation.

$$P_{solution} = p^{\circ}_{toluene} + x_{benzene}(p^{\circ}_{benzene} - p^{\circ}_{toluene}) \text{ ---------- (9.10)}$$

9.7.2 Vapour pressure of binary solution of solids in liquids

When a nonvolatile solute is dissolved in a pure solvent, the vapour pressure of the pure solvent will decrease. In such solutions, the vapour pressure of the solution will depend only on the solvent molecules as the solute is nonvolatile.

For example, when sodium chloride is added to the water, the vapour pressure of the salt solution is lowered. The vapour pressure of the solution is determined by the number of molecules of the solvent present in the surface at any time and is proportional to the mole fraction of the solvent.

Unit 9.indd 44 23-12-2021 17:00:55